

NAAC FILE MANAGEMENT SYSTEM

A Digital Portal for Departmental NAAC Documentation

7 Criteria Forms

File Upload

PDF & Excel Export

HOD Dashboard

PROBLEM STATEMENT

Indian colleges maintain NAAC accreditation files manually across 7 criteria, resulting in scattered documents, missed submissions, and a fragmented review process. There is no unified digital system for departments to upload, track, and submit supporting evidence in a structured, auditable format.

Build a web portal that solves this — criterion by criterion.

7

NAAC Criteria

34+

Required Files

2

User Roles

2

Export Formats

Role 1: Teacher

Role 2: Super User (HOD)

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1. Problem Statement

Indian colleges undergo NAAC accreditation every 5 years. Each department must maintain 34+ files across 7 criteria and submit supporting evidence to the IQAC. Currently, this process is entirely manual — physical files, scattered spreadsheets, and email chains — leading to lost documents, incomplete submissions, and last-minute panic.

Key Pain Points:

No Centralised System	Each department stores files locally with no unified view.
Manual Compilation	HOD spends weeks collecting, verifying and compiling data.
No Progress Tracking	No one can see which departments have submitted which files.
Risk of Data Loss	Paper files and local drives get corrupted or misplaced.
No Structured Format	Documents are not mapped to NAAC criteria, making audits difficult.
No Export to NAAC Format	Final submission requires manual re-entry into NAAC templates.

SOLUTION: Build a web-based NAAC File Management System where teachers can log in, fill structured forms, and upload supporting documents criterion-wise. The HOD (Super User) can review all teachers' submissions, export department data as PDF or Excel, and download a consolidated sheet — ready to submit to NAAC.

2. About NAAC & The 7 Criteria

NAAC (National Assessment and Accreditation Council) is an autonomous body under UGC that assesses and accredits Higher Education Institutions in India. Institutions are graded A++, A+, A, B++, B+, B, C, or D based on a 1000-point framework across 7 criteria.

Code	Criterion Name	Max Marks
C1	Curriculum Aspects	100
C2	Teaching-Learning & Evaluation	350
C3	Research, Innovations & Extension	110
C4	Infrastructure & Learning Resources	100
C5	Student Support & Progression	140
C6	Governance, Leadership & Management	100
C7	Institutional Values & Best Practices	100

Total: 900 marks (remaining 100 are for peer team assessment at institution level)

3. Sub-Criteria Breakdown — All 7 Criteria

Below is the detailed list of sub-criteria under each criterion. Your system must have structured data entry forms for each item.

Criterion 1	Curriculum Aspects	100 marks
1.1	Curricular Planning & Implementation Teachers in BoS/Academic Council/Syndicate (2016-21); New programmes introduced during 2016-21.	
1.2	Academic Flexibility Syllabi of courses relevant to Gender, Environment, Sustainability, Human Values and Professional Ethics; Audit Courses list.	
1.3	Curriculum Enrichment Comprehensive Syllabi of Core/Common/Electives/Practicals/Open Courses 2016-21; List of students undertaking Field Projects/Internships with proof of certificates.	
1.4	Feedback System Structured feedback from students on Syllabi; Analysis and Action Taken Reports (ATRs); Add-on and Value-added Courses with details.	

Criterion 2	Teaching-Learning & Evaluation	350 marks
2.1	Student Enrolment & Profile Nominal Rolls of students admitted 2016-21; List of students in reserved categories (OBC/SC/ST/PH) with Bio-Data and category certificate proof.	
2.2	Catering to Student Diversity List of Permanent/Guest Faculty with full profiles — PhDs, Fellowships, Awards, Recognitions, Projects, Papers, Books; Details of ICT tool usage; Special strategies for slow learners.	
2.3	Teaching-Learning Process Internal Assessment details for last 5 years; Department Minutes Book of grievance redressed; Academic Calendar and Department minutes book with work diary.	
2.4	Teacher Quality Programme/Course Outcomes file; File showing attainment of programme outcomes; Faculty qualifications and research guide recognition.	
2.5	Evaluation Process & Reforms Results file for last 5 years showing split-up of pass percentage with detailed statistics per course.	

Criterion 3 Research, Innovations & Extension		110 marks
3.1	Resource Mobilisation for Research Grants for research projects sponsored by government/non-government agencies with all details (2016-21); Teachers recognised as research guides with orders and scholar details.	
3.2	Innovation Ecosystem Details of Workshops, Seminars, and industry-academia innovative practices hosted by the department.	
3.3	Research Publications & Awards Number of research papers per teacher in UGC-notified journals; Books, book chapters, and papers in National/International proceedings (last 5 years); PhDs awarded per teacher.	
3.4	Extension Activities Extension activities hosted in neighbourhood; Details of students participating in extension activities 2016-21.	
3.5	Collaboration Details of faculty/student exchange, internships, field trips, on-the-job training, and research collaborations during last 5 years.	

Criterion 4 Infrastructure & Learning Resources		100 marks
4.1	Physical Facilities File showing details of classrooms, labs, computing equipment, and research facilities — with photographs, floor plans, and equipment inventory.	
4.2	Library as a Learning Resource Library files to be separately maintained — total holdings, e-resources subscribed, annual usage statistics, library budget utilisation.	

Criterion 5 Student Support & Progression		140 marks
5.1	Student Support Details of students benefitted by government scholarships and fellowships 2016-21; Awards/medals for outstanding performance in Sports/Cultural activities at National/International level with certificate proof.	
5.2	Student Progression Details of placement of outgoing students 2016-21; Student progression to higher education (UG-PG, MA-MPhil-PhD); Students qualifying in NET/SLET/GATE/GMAT/CAT/GRE/TOFEL/UPSC/PSC 2016-21.	
5.3	Student Participation & Activities Details of alumni association meetings held during last 5 years 2016-21 — agenda, minutes, and photographs.	

Criterion 6 Governance, Leadership & Management		100 marks
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6.1	Institutional Vision & Leadership FDP Certificates — Faculty Development Programme attendance and participation records with programme details.
6.2	Strategy Development & Deployment PBAS Files — Performance Based Appraisal System records for all faculty members.
6.3	Faculty Empowerment & Fund Generation Files regarding fund generation activities (internal resource mobilisation or external grants) if any.

Criterion 7 Institutional Values & Best Practices		100 marks
7.1	Institutional Values & Social Responsibilities Files related to Green Initiatives (energy, water, e-waste); Disabled-friendly facilities; Gender Equity activities and programmes.	
7.2	Best Practices Files related to Best Practices and Institutional distinctiveness — minimum 2 best practices with objectives, context, evidence of success, and supporting documents.	

4. What Participants Must Build

Participants must design and develop a fully functional, responsive web application that acts as a centralised NAAC documentation portal. The application must cover all of the following modules:

01 Authentication System

Secure login/registration for 2 roles: Teacher and Super User (HOD). Each role sees a different dashboard with different permissions.

02 Criterion-Wise Data Entry Forms

Structured online forms for all 7 NAAC criteria and their sub-criteria. Each form field maps to a specific NAAC requirement. Include file upload for supporting documents (PDF, JPG, DOCX).

03 Document Upload & Management

A file manager per criterion where teachers can upload, view, rename, and delete supporting documents. Show upload status: Pending / Uploaded / Verified.

04 Progress Dashboard

Visual dashboard showing completion percentage per criterion, overall department progress, and pending items. Use charts/progress bars.

05 PDF & Excel Export

Teachers can export their own data as a formatted PDF or Excel. The HOD (Super User) can export all teachers' data combined.

06 HOD Consolidated Download

HOD (Super User) views all teachers' submissions and downloads a single consolidated Excel sheet — ready to submit to NAAC.

07 Notifications & Reminders

In-app notifications reminding teachers of pending uploads and approaching NAAC submission deadlines.

08 Audit Trail / Activity Log

Log all actions (who uploaded what and when). HOD can view the full activity history.

5. Project Goals & Expectations

The judging panel expects a practical, polished, and genuinely usable solution:

Functional Completeness

- All 7 criteria must have working data entry forms.
- Upload, view, edit, and delete operations must work.
- Both PDF and Excel export must be functional.
- HOD consolidated sheet must be downloadable.

Data Integrity

- Form validation on all fields (required fields, date formats, numeric ranges).
- File type validation on uploads (accept only PDF, DOCX, JPG, PNG).
- No data loss between sessions — data must persist in a database.

User Experience

- Clean, intuitive UI that a non-technical teacher can use without training.
- Responsive design — must work on both desktop and tablet.
- Clear progress indicators, success messages, and error alerts.

Security

- Role-based access control enforced on every page.
- Teacher A must NOT be able to view or edit Teacher B's data.
- Secure file storage — no direct URL access to uploaded files.

Report Quality

- PDF reports must be neatly formatted with department/college branding.
- Excel sheets must use proper headers and structure.
- HOD consolidated sheet must be ready for NAAC submission.

6. How the Website Works — Functional Flow

STEP 1	Teacher Registers / Logs In Each teacher is given a unique login. On first login, the teacher completes their profile (name, department, subjects taught, etc.).
STEP 2	Select a Criterion to Work On The dashboard shows all 7 criteria with completion badges. The teacher clicks any criterion to open its sub-criteria forms.
STEP 3	Fill Data Entry Forms Structured forms are shown for each sub-criterion. The teacher fills text fields, numeric inputs, date pickers, and dropdowns — then clicks Save.
STEP 4	Upload Supporting Documents After filling the form, the teacher uploads supporting files (PDFs, certificates, photos). Each upload is tagged to the exact sub-criterion.
STEP 5	Teacher Exports Own Report When data entry is complete, the teacher clicks Export to download their own data as a formatted PDF or Excel sheet, criterion-wise.
STEP 6	HOD Reviews All Submissions The HOD (Super User) logs in and can see all teachers' submissions. HOD marks each as Verified or Needs Revision with comments.
STEP 7	HOD Downloads Consolidated Sheet The HOD clicks Download All. The system generates a single Excel workbook with all teachers' data — ready for NAAC submission.

7. Data Entry Forms — Criterion-Wise Field Guide

Below are the required form fields for each criterion. Participants must implement all fields with proper input types, validation, and labels.

Criterion 1 — Curriculum Aspects

Field Name	Input Type	Notes / Validation
Programme Name	Text	e.g., B.E. Computer Engineering
Academic Year	Dropdown	2016-17 to 2020-21
Course Type	Dropdown	Core / Elective / Practical / Open / Audit
New Programme Introduced?	Yes/No Radio	If Yes, attach approval document
BoS Member Name	Text + Date	Name and date of nomination to Board of Studies
Field Project Title	Text	Student project/internship topic
Internship Organisation	Text	Company or institution name
Certificate	File Upload	PDF or image of certificate
Value Added Course Name	Text	Course name and number of hours
Feedback Collected?	Yes/No + File Upload	Upload ATR (Action Taken Report) document

Criterion 2 — Teaching-Learning & Evaluation

Field Name	Input Type	Notes / Validation
Student Full Name	Text	As per college records
Roll Number	Text	Unique student identifier
Category	Dropdown	General / OBC / SC / ST / PH
Year of Admission	Year Picker	2016 to 2021
Faculty Name	Text	Full name of faculty member
Designation	Dropdown	Professor / Asst. Professor / Guest Faculty
Highest Qualification	Dropdown	UG / PG / M.Phil / Ph.D
Number of Publications	Number	Journal papers in last 5 years

ICT Tools Used	Multi-select	PPT / LMS / Videos / CDs / Software
Internal Assessment Marks	Number	Per student per course
IA Document	File Upload	Scanned IA sheets as PDF
Pass Percentage (Year-wise)	Decimal	Per course per academic year
Result Statistics File	File Upload	Upload consolidated result sheet
Special Strategies for Slow Learners	Textarea	Describe methods used

Criterion 3 — Research, Innovations & Extension

Field Name	Input Type	Notes / Validation
Research Project Title	Text	Full title of funded project
Principal Investigator	Text	Faculty member name
Funding Agency	Text	Name of Govt. / Non-Govt. body
Grant Amount (INR)	Number	Total grant sanctioned
Project Start Date	Date Picker	—
Project End Date	Date Picker	—
Journal Paper Title	Text	Full title of research paper
Journal Name	Text	Name of publication journal
UGC Listed?	Yes/No Radio	Is the journal in UGC-CARE list?
ISSN Number	Text	International Standard Serial Number
Book / Chapter Title	Text	For books and book chapters
Publisher Name	Text	Name of publishing house
Year of Publication	Year Picker	—
PhD Scholar Name	Text	Name of research scholar
PhD Awarded Year	Year Picker	Year of PhD conferment
Workshop/Seminar Name	Text	Title of event organised/attended
Event Date	Date Picker	—
Brochure / Certificate	File Upload	PDF of brochure or certificate
Extension Activity Name	Text	Name of outreach/community activity
Number of Students Participated	Number	Count of student participants

Criterion 4 — Infrastructure & Learning Resources

Field Name	Input Type	Notes / Validation
Facility Type	<i>Dropdown</i>	<i>Classroom / Lab / Computer Lab / Research Facility</i>
Room Number / Name	<i>Text</i>	<i>Identifier for the facility</i>
Area (sq. ft.)	<i>Number</i>	<i>Carpet area of the space</i>
Seating Capacity	<i>Number</i>	—
Number of Systems (for computer labs)	<i>Number</i>	—
Equipment Name	<i>Text</i>	<i>Individual equipment/instrument</i>
Equipment Make & Model	<i>Text</i>	—
Year of Purchase	<i>Year Picker</i>	—
Cost (INR)	<i>Number</i>	<i>Purchase cost</i>
Facility Photograph	<i>File Upload</i>	<i>JPG or PNG — minimum 2 photos</i>

Criterion 5 — Student Support & Progression

Field Name	Input Type	Notes / Validation
Scholarship Name	<i>Text</i>	<i>Name of the scholarship scheme</i>
Awarding Body	<i>Text</i>	<i>Govt. body or institution giving the award</i>
Beneficiary Student Name	<i>Text</i>	—
Category	<i>Dropdown</i>	<i>SC / ST / OBC / Minority / Merit</i>
Amount (INR)	<i>Number</i>	<i>Annual scholarship amount</i>
Academic Year	<i>Dropdown</i>	<i>2016-17 to 2020-21</i>
Placement Company	<i>Text</i>	<i>Company where student was placed</i>
Job Role	<i>Text</i>	<i>Designation / role offered</i>
Package (LPA)	<i>Decimal</i>	<i>Annual package in lakhs</i>
Higher Education Institution	<i>Text</i>	<i>UG to PG progression destination</i>
Programme Pursued	<i>Text</i>	<i>e.g., MBA, M.Sc, LLB</i>
Competitive Exam	<i>Dropdown</i>	<i>NET / SLET / GATE / UPSC / PSC / CAT / GRE / TOFEL / GMAT</i>
Exam Year	<i>Year Picker</i>	—
Rank / Score	<i>Text</i>	—
Award / Medal Name	<i>Text</i>	<i>Sports or cultural achievement</i>

Level	Dropdown	State / National / International
Certificate	File Upload	Scanned certificate
Alumni Event Date	Date Picker	—
Alumni Meeting Minutes	File Upload	PDF of meeting minutes

Criterion 6 — Governance, Leadership & Management

Field Name	Input Type	Notes / Validation
FDP Programme Name	Text	Full title of Faculty Development Programme
Organising Institution	Text	Name of organising body
Duration (Days)	Number	Total days of the programme
Faculty Member Name	Text	Attendee name
Date of Attendance	Date Picker	—
Certificate	File Upload	FDP completion certificate
PBAS Score	Decimal	Performance-Based Appraisal Score
Assessment Year	Year Picker	—
Fund Generation Activity	Text	Name of revenue-generating activity
Amount Generated (INR)	Number	Total amount collected/generated
Source of Funds	Text	How funds were raised
Supporting Document	File Upload	Receipts, sanction letters, etc.

Criterion 7 — Institutional Values & Best Practices

Field Name	Input Type	Notes / Validation
Best Practice Title	Text	Name of the best practice
Objective	Textarea	Goals and aims of the practice
Context	Textarea	Background and need for the practice
Practice Description	Textarea	Step-by-step description of implementation
Evidence of Success	Textarea	Outcomes, impact, numbers achieved
Problems Encountered	Textarea	Challenges faced during implementation
Resources Required	Textarea	Human, financial, physical resources
Evidence Document	File Upload	PDF report or photographs
Green Initiative Name	Text	Name of eco-friendly initiative

Gender Equity Activity	<i>Text + Date</i>	<i>Activity name and date conducted</i>
Disabled-Friendly Feature	<i>Text</i>	<i>e.g., Ramps, Braille, Special washrooms</i>
Supporting Document	<i>File Upload</i>	<i>Photo or official document</i>

8. Export Features — PDF & Excel Reports

The system must support two types of exports, generated dynamically from the database. No manual editing should be required.

8A. PDF Report (Per Teacher)

- Department/college name and teacher name on the cover page.
- Separate section for each criterion with heading and marks.
- All form data displayed in clean table format per sub-criterion.
- List of uploaded documents with file names and upload dates.
- Summary table at the end showing completion status per criterion.
- Page numbers, date of export, and academic year in the footer.
- Generated using server-side libraries (ReportLab / jsPDF / WeasyPrint).

8B. Excel Report (Per Teacher)

- One worksheet per criterion (7 sheets total).
- Row 1: Criterion name and marks as merged header cell.
- Row 2: Column headers (Field Name, Value, Supporting Document, Status).
- Data rows with alternating colours for readability.
- Frozen header row for easy scrolling.
- A Summary sheet as the first tab showing overall progress.
- Generated using openpyxl / ExcelJS / SheetJS.

9. HOD Dashboard & Consolidated Submission

The HOD (Super User) has a special dashboard that aggregates data from all teachers in the department. This is the most critical feature for actual NAAC submission.

HOD Dashboard Must Show:

- Total teachers registered vs. teachers that have started data entry.
- Criterion-wise completion status for each teacher.
- List of teachers with pending submissions and last activity date.
- Quick links to view any teacher's data in read-only mode.
- Option to send reminders to specific teachers.
- Verification status per teacher per criterion (Pending / Verified / Needs Revision).

Consolidated Excel Sheet (NAAC Submission Format):

The HOD clicks Download All Teachers. The system generates a single Excel workbook (.xlsx): Tab 1 — Summary: All teachers, all criteria, completion %, verification status. Tabs 2-8 — One tab per criterion: All teachers' data for that criterion in rows. Tab 9 — Document Index: Complete list of all uploaded files across all teachers. This workbook can be directly submitted to NAAC or to the institution's central IQAC without any manual reformatting.

Teacher-Level Download:

Each teacher can also download only their own data as a standalone PDF or Excel to review and cross-check before the HOD locks the final submission.

10. User Roles & Access Control

The system has exactly 2 user roles. Each role has a distinct dashboard and clearly defined permissions.

Role 1 Teacher

- Register with name, department, and credentials.
- Fill data entry forms for all 7 criteria.
- Upload supporting documents per sub-criterion.
- View and edit only their own data.
- Track their own progress per criterion.
- Download PDF / Excel of their own data.
- Cannot view or edit other teachers' data.
- Cannot access the HOD dashboard.

Role 2 Super User (HOD)

- View all teachers' submissions in read-only mode.
- Mark submissions as Verified or Needs Revision.
- Add review comments per criterion per teacher.
- Send reminders to individual teachers.
- Download consolidated Excel (all teachers, all criteria).
- View audit logs — who did what and when.
- Manage teacher accounts (approve, deactivate).
- Cannot fill data forms (those belong to teachers).

Security Note: Role-based access control must be enforced on every route and API endpoint. A teacher must never be able to access another teacher's data or the HOD dashboard — even by directly typing a URL. All file uploads must be stored securely with no public direct access.

11. Technical Stack Recommendations

Participants may use any technology stack. Below are recommended options — not a restriction.

Layer	Recommended Options
Frontend	React.js / Next.js / Vue.js / HTML + Bootstrap / Tailwind CSS / Angular
Backend	Node.js + Express / Python + Django / Flask / PHP + Laravel / Java Spring Boot
Database	MySQL / PostgreSQL (recommended — relational) / MongoDB (NoSQL) / SQLite (dev only)
PDF Gen	Python: ReportLab, WeasyPrint / Node.js: Puppeteer, jsPDF / PHP: DOMPDF
Excel Gen	Python: openpyxl, XlsxWriter / Node.js: ExcelJS, SheetJS / PHP: PhpSpreadsheet
File Store	Local filesystem (dev) / AWS S3 / Cloudinary (prod) / Firebase Storage
Auth	JWT (JSON Web Tokens) / Session-based auth / OAuth2 (Google login)

Tip: Use a relational database (MySQL / PostgreSQL) because NAAC data has clear relationships — Teacher > Criterion > Sub-criterion > Form Fields > Uploaded Files. A well-designed schema will make report generation significantly easier.

12. Judging Criteria & Deliverables

Parameter	Description
Functionality (35 marks)	All 7 criteria forms working; Upload working; PDF & Excel export working; HOD consolidated download working.
UI / UX Design (20 marks)	Clean, professional interface; Responsive on desktop and tablet; Intuitive navigation; Clear role separation.
Code Quality (15 marks)	Well-structured code; Meaningful variable names; Separation of concerns; Commented where necessary.
Database Design (10 marks)	Logical schema; Proper use of foreign keys; No redundant data; Efficient queries.
Security (10 marks)	Role-based access control on every route; Input validation; Secure file upload; Data isolation between teachers.
Report Quality (10 marks)	PDF neatly formatted; Excel structured and ready for NAAC submission; HOD consolidated sheet well-organised.

Deliverables to Submit:

- Working web application (deployed link or localhost demo with video walkthrough).
- Source code (GitHub repository link or zipped folder).
- Database schema file (.sql or entity-relationship diagram).
- README file with clear setup and installation instructions.
- A 5-minute demo video or live presentation to judges.
- A sample exported PDF report and Excel sheet from the system.

13. Submission Checklist

Use this checklist before your final submission to ensure nothing is missed.

Authentication & Roles

- ☐ Login / Registration working for both Teacher and HOD (Super User)
- ☐ Role-based dashboard shown correctly after login
- ☐ Teacher A cannot access Teacher B's data
- ☐ HOD can view all teachers but cannot edit their forms
- ☐ Password hashing and secure session management implemented

Data Entry Forms

- ☐ All 7 criterion forms are present and working
- ☐ All fields have proper input types and validation
- ☐ Forms save data to the database correctly
- ☐ Edit / Update functionality works
- ☐ Delete with confirmation prompt

Document Upload

- ☐ File upload works for all form sections
- ☐ Only allowed file types accepted (PDF, DOCX, JPG, PNG)
- ☐ Uploaded files can be viewed and downloaded
- ☐ Files are tagged to the correct criterion and sub-criterion

Progress Dashboard

- ☐ Completion percentage shown per criterion for teacher
- ☐ HOD can see all teachers' progress at a glance
- ☐ Pending items clearly highlighted
- ☐ Charts or progress bars are present

Export Features

- ☐ Teacher PDF export is formatted and complete
- ☐ Teacher Excel export has 7 criterion sheets + Summary tab
- ☐ HOD consolidated Excel includes all teachers' data
- ☐ Consolidated sheet is structured for NAAC submission

HOD Panel

- ☐ All teachers visible in HOD dashboard
- ☐ Consolidated download button works
- ☐ HOD can mark submissions as Verified / Needs Revision
- ☐ Audit log / activity history is present
- ☐ Reminder notification can be sent to individual teachers

Good luck to all participants! Remember — the best system is one that a real teacher can use on the day of NAAC inspection without any guidance. Build for the user, not just for the judge. We look forward to seeing your solutions!